

1) Definitions of Econometrics:- 2021

The word "Econometrics" first introduced in 1926 by Ranges Frisch has various definition. i- Econometrics may be defined as the social science in which the tools of economic theory, ~~math~~ mathematics, and statistical inference are applied to the analysis of economic phenomena. [Arthur S. Goldberger] Econometrics may be defined as the quantitative analysis of actual economic phenomena.

2) Scope of econometric:-

The scope of econometrics is much broader than the "economic measurements."

3- ^{scope} uses of econometrics:-

- There are three major uses of econometrics.
- i- Describing economic reality [modeling and estimating].
 - ii- Testing hypothesis about economic theory [Analysis].
 - iii- Forecasting:- Econometric models are used to forecast economic variable such as GDP, inflation and unemployment.
 - iv- Cost-benefit analysis:- Econometrics is used to evaluate the costs and benefits of projects or policies.
 - v- Agricultural economics:- Econometrics is used to study agricultural production, prices and trade.
 - vi- Real estate economics:- Econometrics is used to study real estate markets, prices and investment.
 - vii- International trade:- Econometrics is used to study international trade patterns and the impact of trade policies.

4- Division of Econometrics:-

- Econometrics may be divided into two types:
- theoretical econometrics and applied econometrics.
- i- Theoretical Econometrics:- Theoretical econometrics involves the development of appropriate methods for the measurements of economic phenomena (relationship).
 - ii- Applied econometrics:- Applied econometrics involves the application of the techniques of theoretical econometrics for the analysis and forecasting economic relationships.

5- Econometrics modeling:-

The term "model" appears very frequently in econometric literature. An economic model relates to a set of structures defined by relationships b/w economic variables.

→ An econometric model is an economic model that includes the variables from economics, business and other social sciences.

→ The model often used for policy analysis and forecasting.

→ Econometrics models are expressed in equation form.

6- Methodology of Econometrics:-

Major ingredients of Econometric modelling:-

i- Behavioural relationship.

ii- Definitions or identities.

iii- Equilibrium relations.

iv- Technological Relationships.

v- Institutional or Historical Relations. 2016-ii

6 methodology of Econometrics:- SEVF

The traditional econometric methodology involves the following four steps.

- i- specification
- ii- Estimation
- iii- verification or Evaluation
- iv- Forecasting.

- The first step is the specification of econometric model of the theory that has been done with the help of specifying the mathematical model of the theory.

ii- The 2nd stage involves the numerical estimates of the parameters of the model. This has been done with the help of regression analysis.

iii - Verification or Evaluation:

The third stage in econometric research involves the evaluation of the estimated ~~of the estimated~~ model on the basis of the ~~(i)~~ Prior theoretical criteria ii) Statistical criteria iii) Econometric criteria.

iv - Forecasting:-

This refers to the ability of the model to accurately predict future values of the dependent variable based on known or expected future values of explanatory variables.

7. Function of Econometrics. 2017-iv-2021

Econometrics has three closely interrelated functions.

i - First: The first function is to test economic theories or hypothesis.

ii - Second:- The 2nd function is to provide numerical estimates of the co-efficients of economic relationships.

iii - Third:- The third function is the forecasting of economic events.

8. ^{scope} need of econometrics as a separate discipline.

Analysis of economic data: Econometrics provides tools to analyze and interpret economic data, which is essential for understanding economic phenomena and making informed decision.

i - Testing Economic theories: Econometric allows for the testing of economic theories and models.

ii - Policy evaluation: Econometrics helps evaluate the effectiveness of economic policies and models.

iii - Forecasting and prediction: Econometrics enables economists to forecast economic trends and predict future outcomes.

9- Diff. Prediction and forecasting:

Prediction:-

Prediction is the process of using statistical models, machine learning algorithms, or other techniques to estimate or forecast outcomes, values or probabilities for unknown or future data.

- It involves making an estimate about a future event.
- Prediction can be applied to various domains, such as classification, regression, clustering and density estimation.
- The goal of prediction is to make accurate estimates for unseen or future data.

Forecasting:-

Forecasting is a specific type of prediction that focuses on estimating future values or outcomes.

- It involves using historical data or statistical models to predict future values.
- Forecasting is commonly used in economics, finance and business to predict future events.
- The goal of forecasting is to provide accurate and reliable estimates of future values or trends.

10- Under what reason an error term is introduced in econometric model?

An error term is introduced in an econometric model for the following reasons:

- i- Uncertainty: The error term is used to indicate the uncertainty in the model.
- ii- Lack of perfect goodness of fit: The error term accounts for the difference b/w the actual observed results and the theoretical value of the model.

iii - margin of error:

The error term represents the sum of the deviations within the regression line.

iv - Incompleteness:- The error term is a residual variable produced by a statistical or mathematical model, which is created when the model doesn't fully represent the actual relationship b/w the independent variables and dependent variables.

v - Residual:- The error term can be used to represent the residual or remainder term in a model.

11- Properties of OLS estimators:

The OLS estimators are best, linear and unbiased estimators.

i - Linearity:- The OLS estimators are linear as they can be expressed as a linear function of random variable (Y_i).

ii - unbiasedness:- The OLS estimators are unbiased as the average or expected values are equal to the true values. $E(\hat{\beta}) = \beta$ (ii) $E(\hat{\alpha}) = \alpha$.

iii - variance: $V(\hat{\beta}) = \sigma^2 (\sum x_i^2)^{-1}$. $V(\hat{\alpha}) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum x_i^2} \right]$, $Cov(\hat{\alpha}, \hat{\beta}) = \frac{-\bar{x} \sigma^2}{\sum x_i^2}$

iv - minimum variance:

$$V(\beta^*) = V(\hat{\beta}) + \sigma^2 \sum (c_i - w_i)^2 \Rightarrow V(\hat{\beta}) < V(\beta^*)$$

Thus $\hat{\beta}$ is a BLUE of β .

$$V(\alpha^*) = V(\hat{\alpha}) + \sigma^2 \bar{x} \sum (c_i - w_i)^2 \Rightarrow V(\hat{\alpha}) < V(\alpha^*) \text{ Thus } \hat{\alpha} \text{ is BLUE of } \alpha$$

v - Sampling distⁿ:-

$$\hat{\beta} \sim N[\beta, \sigma^2 (\sum x_i^2)^{-1}]$$

$$\hat{\alpha} \sim N[\alpha, \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum x_i^2} \right]]$$

12- Restricted least square method:-

The restricted least square method is a statistical technique used to estimate parameters in a linear regression model. The key points of the restricted least square method are:

- i- Restricted Regression:- It puts linear constraints on the coefficients in the model.
- i- Geometry:- Geometrically, OLS regression is the orthogonal projection of the observed response (Y) onto the column space of the design matrix.
- i- Linear algebra:- This can be done starting from Y or starting from the OLS solution.
- i- Equality constraints:- RLS can handle equality constraints.
- i- Inequality constraints:- RLS can't handle inequality constraints.

13- Sampling distⁿ of regression SS.

The sampling distⁿ of regression SS is a Chi-Square distⁿ with $(K-1)$ d.o.f, where K is the number of independent variables in the regression model.

- > The expected value of the regression SS is $(K-1)$ times the variance of the error term σ^2 .
- > The variance of the regression SS is $2(K-1)$ times the variance of the error term σ^2 .
- > The sampling distⁿ of the regression SS is asymptotically Chi-squared, as the sample size increases.

✓gn GLRM in deviation form under $H_0: \beta = 0$

$$Y = X\beta + u$$

$$\hat{Y} = X\hat{\beta}$$

$$\text{Reg SS} = \hat{\beta}' X' Y$$

$$= \hat{\beta}' X' Y$$

$$= [(X'X)^{-1} X' Y]' X' Y = Y' X (X'X)^{-1} X' Y$$

$$= (X\beta + u)' X (X'X)^{-1} X' (X\beta + u)$$

$u' X (X'X)^{-1} X' u$ which is quadratic form in normal vector $u \sim N(0, \sigma^2 I)$

Therefore, Reg SS $\sim \sigma^2 \chi^2$ distⁿ with

$$\text{d.f} = \text{Rank} \left[\frac{X}{A} \frac{(X'X)^{-1} X'}{\beta} \right] = \text{tr} \left[\frac{X}{A} (X'X)^{-1} X' \frac{1}{\beta} \right]$$

$$\text{d.f} = \text{tr}(I_K) \Rightarrow \text{d.f} = K$$

$$\text{So, Reg SS} = \sigma^2 \chi^2_{K-1}$$

14- Gauss markov's theorem and importance:

The Gauss-Markov theorem states that if certain conditions are met, the OLS estimator of the coefficients of a linear regression model is the BLUE. The conditions are as follows.

Linearity: The parameters being estimated using the OLS method must be linear.

Randomness: The data must have been randomly sampled from the population.

Exogeneity: The regression aren't correlated with the error term.

Homoscedasticity: The variance of the error term is constant.

15- Chow test:- GLRM

The Chow test is a statistical test used to determine whether the coefficients in a regression model are equal across different groups or subsamples.

The Chow test in GLRM involves the following steps:

1. Divide the data into two or more subsamples.

2. Estimate the GLRM model separately for each subsample.

3. Compute the log-likelihood functions for each subsample.

4. Compute the Chow test statistics.

5. Compare the test statistics to a critical value from a chi-squared distribution.

$$F = \frac{(SSE_R - SSE_{UR}) / K + 1}{SSE_{UR} / (n_1 + n_2 - 2K - 2)}$$

{ $\alpha_1, \beta_1, \alpha_2, \beta_2$

{ $\alpha_1, \beta_1, \alpha_2, \beta_2$

{ (3) log-likelihood

16 - Assumption of MLE in GLRM:-

- i- x_i 's are fixed variables.
- ii- x_i 's are linearly uncorrelated i.e. there is no relationship b/w x_i 's.
- iii- Error terms are distributed independently with mean zero and variance same.
i.e. $E(\epsilon_i) = 0$, $\text{var}(\epsilon_i) = \sigma^2$.
- iv- Error term follows Normal distⁿ. with characteristics explained in assumption (iii) i.e.
 $\epsilon_i \sim NID(0, \sigma^2)$.

17 - Test statistics to test some regression coefficients in GL Regression:-

$$t = \frac{(\hat{\beta}_{1j} - \hat{\beta}_{2j}) - (\beta_{1j} - \beta_{2j})}{\sigma_p^2 [b_{1jj} + b_{2jj}]^{1/2}}$$

follows t -distⁿ with $(n_1 + n_2 - 2k - 2)$ d.f.

$$F = \frac{Q_2 / k - H}{SSE / n - k - 1}$$

2. Properties of MLE of error variance in GLS.

- i- Efficient (minimum variance)
- i- Consistency
- i- Asymptotic normality.

19. Stepwise regression:-

Stepwise regression is a statistical technique used in regression analysis to select the most relevant independent variables for a model. It involves a step-by-step procedure to add or remove variables from the model based on their statistical significance.

- There are two common types of stepwise regression:
- 1- Forward selection: Start with no variables and add them one-by-one.
 - 2- Backward elimination: Start with all variables and remove them one-by-one.

Stepwise regression helps to:

- i- identify the most important predictors.
- ii- Reduce multicollinearity.
- iii- Improve model interpretability.
- iv- Avoid overfitting.

20- Ridge Regression:-

Ridge regression is a technique for analyzing multiple regression data that suffer from multicollinearity. When multicollinearity occurs, least square estimates are unbiased, but their variances are large, so they may be far from the true value.

→ Ridge regression is commonly used in situations where:

- i- There are many correlated predictors.
- ii- The data is noisy or has outliers.
- iii- model interpretability is not a priority.

+ Benefits:- IRI

- i- improved model stability.
- ii- Reduced variance.
- iii- Increased bias.

21- Extraneous information in linear regression and uses.

Extraneous information in linear regression refers to data or variables that are not relevant to the relationship b/w the independent and dependent variables. This information can:

- i- Increase the risk of overfitting.
- ii- Reduce model interpretability.
- iii- Inflate variance.

By the use of extraneous information, when we estimate the parameters, there are restricted estimators. Restricted estimators are more precise as compare to unrestricted estimators.

$$\text{i.e. } V(\hat{\beta}_R) < V(\hat{\beta}).$$

22. Sampling Distⁿ of error SS.

$$\text{AS } e = y - \hat{y} \quad \text{or } e_i = y_i - \hat{y}_i$$

$$e \sim N(0, \sigma^2 I) \quad , \quad e_i \sim N(0, \sigma^2)$$

So: $e'e \sim \chi^2 \sigma^2$ distⁿ with d.o.f determined as

expected value of χ^2 .

$$E(e'e) = E(\sigma^2 \chi^2) = \sigma^2 E(\chi^2)$$

$$E(e'e) = (n-k-1) \sigma^2$$

Thus $e'e = \chi^2 \sigma^2$ with d.o.f = $n-k-1$.